

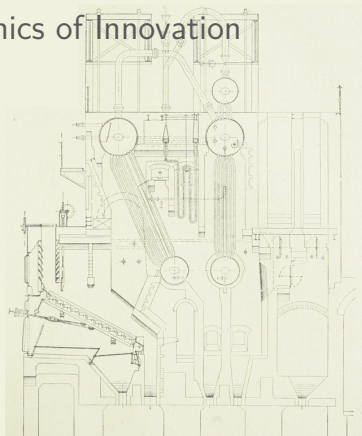
Engineering Germany's Industrial Success

Professional Societies and the Social Dynamics of Innovation

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September 2026



Innovation and Location

Innovation concentrates in a few places — and it stays there.

- Patents cite other patents from the same region far more often (Jaffe et al. 1993)
- Once a place achieves greater innovation, it remains so for decades

Where do inventive places come from?

How did Germany catch up with Britain as an industrial power?

- Between 1850 and 1914, Germany went from an agrarian society to the world's second-largest industrial economy
- Standard explanations: institutions (Acemoglu et al. 2011), coal (Fernihough and O'Rourke 2021), universities (Dittmar and Meisenzahl 2022)
- But universities produced *scientists*. **Who built the machines?**

Kelly et al. (2023): Britain's industrialization rested on *artisan mechanical skill*, not academic science

→ *Germany's approach was organizational*. The German Engineering Society (VDI) brought engineers together, and the staggered rollout of its chapters makes the effect measurable.

This Paper: Organizing Practical Knowledge and Innovation

Question: Did the local chapters of the German Engineering Society cause innovation to rise where they opened?

Concept: Localized knowledge spillovers (Jaffe et al. 1993), organized through a voluntary association (Putnam et al. 1993)

Data:

- Digitized 1900 and 1925 membership directories (43,000) and 50 chapter territories
- All high-value patents in Imperial Germany, 1877–1918, geocoded (Streb et al. 2006)

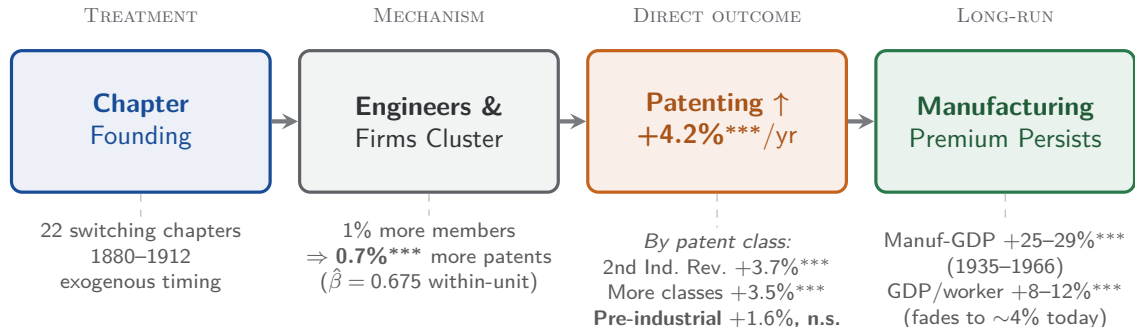
Identification:

- Staggered founding of 22 chapters in the unified Imperial Germany
- Heterogeneity-robust difference-in-differences (Callaway and Sant'Anna 2021)

Main results:

- Chapter founding raises high-value patenting by **4.2%** per year (± 15 yr window)
- Concentrated in 2nd Industrial Revolution technologies (3.7%), flat pre-industrial placebo
- Chapters amplified existing local specialization rather than redirecting it

From Chapter Founding to Long-Run Industrial Strength



Related Literature

Innovation and human capital formation in industrializing Germany

Dittmar and Meisenzahl (2022); Cinnirella and Streb (2017); Donges et al. (2023)

- ▷ Universities, literacy, legal institutions — but not *professional organization*.

This paper studies the engineering association as a distinct channel for innovation

Knowledge spillovers and the social organization of innovation

Jaffe et al. (1993); Satyanath et al. (2017); Cinnirella, Hornung, et al. (2025)

- ▷ Knowledge spillovers are local, and voluntary associations are the networks that carry them.

This paper turns that correlation into a causal estimate: the Engineering Society's staggered rollout identifies what organizing practical knowledge did to innovation

The role of engineers for economic development

Maloney and Valencia Caicedo (2022); Hanlon (2025); Gispén (1989)

- ▷ Maloney: cross-country; Hanlon: Britain, patent-level. Gispén: sociological history of engineers.

This paper provides within-country quasi-experimental evidence on engineering societies for Germany

Outline

- I. Historical Background** From unification to a national network of chapters
- II. Data** Digitized directories and geocoded high-value patents
- III. Empirical Strategy** Staggered chapter founding under one patent law
- IV. Results** The effect on patenting, and where it does not appear
- V. Mechanism** Which places gained, in which technologies, through whom
- VI. Long Run** Whether the advantage outlived the patent window
- VII. Conclusion**

Outline

I. Historical Background

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VII. Conclusion

Germany's transformation and its engineers

- Napoleonic defeat exposed a fragmented, agrarian economy decades behind Britain
- The state answered with schools, unification, and a patent law
- The engineers answered with a society of their own, organized in local chapters

→ The chapter is the treatment in this paper

From Napoleonic Defeat to Industrial Catch-Up

The problem: Napoleonic occupation (1806–13) exposed Germany's technological backwardness. Fragmented, agrarian, and decades behind Britain.

The institutional response:

- ▷ **Polytechnic schools** (from 1825): Eleven institutions by 1880, upgraded to technical universities by 1899. Increasingly theoretical — graduates trained in science, not the shop floor.
- ▷ **Engineering schools** (from 1880s): 29 state schools by 1900. Practical, hands-on training in mechanical engineering. Produced the majority of Germany's working engineers (Gispen 1989).
- ▷ **Political and legal unification:** national unification (1871) and a unified patent law (1877) created the framework for a national innovation system.

The German Engineering Society

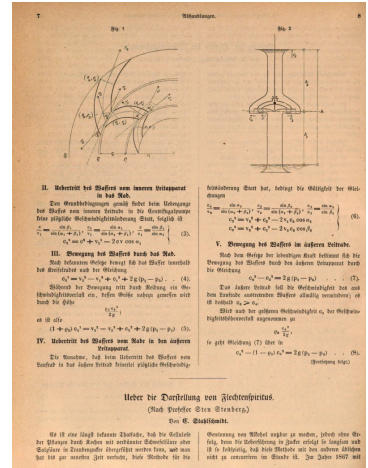
Founded 1856 by 23 young engineers in Alexisbad (Harz). Purpose: unite engineers across state boundaries through local chapters and a technical journal.

Growth:

- 1856: 5 local chapters, 172 members
- 1880: 25 chapters, 4,600 members
- 1900: 40 chapters, 15,500 members

Policy influence:

- Shaped the 1877 Patent Law and lobbied over inventor vs. employer ownership rights (1891)
- Drove engineering school curricula toward practical training through a nationwide standards committee whose guidelines became government policy (Gispert 1989)



Sample page, Magazine 1870

The Engineering Society in Practice

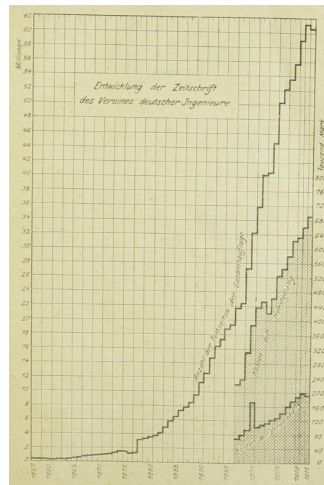
The technical journal:

- Weekly publication distributed to every member across the Empire
- Factory reports, patent digests, technical drawings — a shared knowledge base independent of geography

What local chapters did:

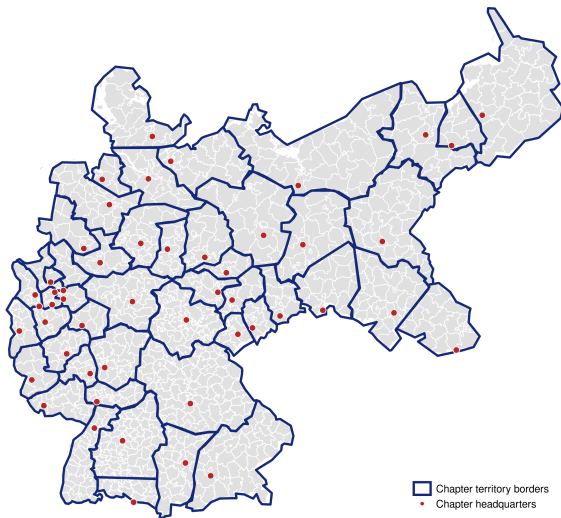
- Monthly meetings: technical lectures, factory visits, machinery demonstrations
- Knowledge exchange between university-trained and shop-floor engineers
- Leadership dominated by factory owners, not professors (Gispen 1989)

→ Chapters were the *local infrastructure* between formal education and the factory floor



Distribution of the Technical Journal

Local Chapter Headquarters Across Germany



Local chapter headquarters and 1925 statistical unit boundaries

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Three sources, one geography

- Membership directories → the engineers
 - Chapter yearbooks → the territories
 - Streb–Baten corpus → high-value patents
- Engineers and patents already cluster together

Data Sources: Historical and Modern

Treatment & membership

- **Chapter panel:** 49 chapters $\times$ 4 cross-sections (1882, 1900, 1912, 1925). Founding dates, membership counts, territories.
- **Membership directories** (transcribed): 15,500 individuals (1900) and 27,200 (1925). Name, city, professional title, employer.

Outcomes

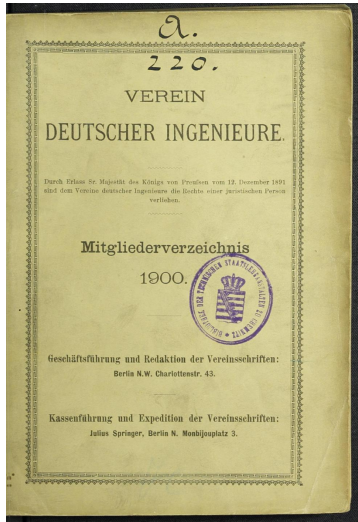
- **High-value patents:** 39,360 patents renewed ≥ 10 years, 1877–1918 (Streb et al. 2006). 89 *Patentklassen* classified into engineering, chemical, mechanical, electrical, transport sectors.
- **Engineer–patent fuzzy match:** ~ 650 person-level links (residence + name).
- **Long-run:** Peters (2022) district-level manufacturing GDP and employment 1933–1966; (BBSR 2019).

Controls and geography

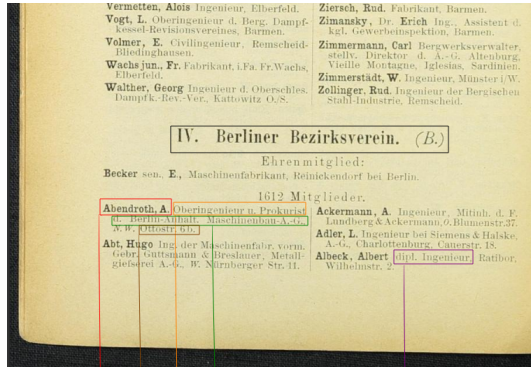
- **Population:** Rösel (2023) GPOP 1871–1950.
- **Railroad network:** Cederman et al. (2025), mapped to county borders 1834–1922.
- **Engineering schools:** 11 Technical Universities, 29 engineering schools (placebo design).
- **Shapefiles:** 1900 statistical units (1,020) and 1882 Donges counties (881), for two-geography robustness.

Geographic units: *Historical:* Stat. unit (1,020) $\subset$ County (1,005) $\subset$ State (22). 50 chapter territories.
Modern: Municipality (10,785) $\subset$ County (400) $\subset$ State (16).

Engineering Society Membership Directory 1900



Membership Directory 1900



Street

Last Name: Abendroth
First Name: A.

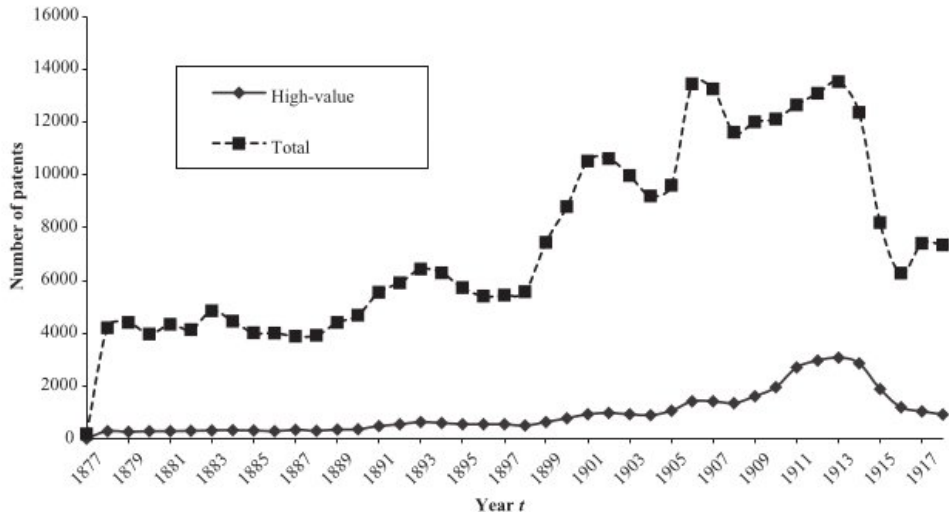
Job Position: Chief Engineer
& Procurator

Employer: Berlin-Anhalt. Maschinenbau A.G.

Academic Title: Diploma Ingenieur

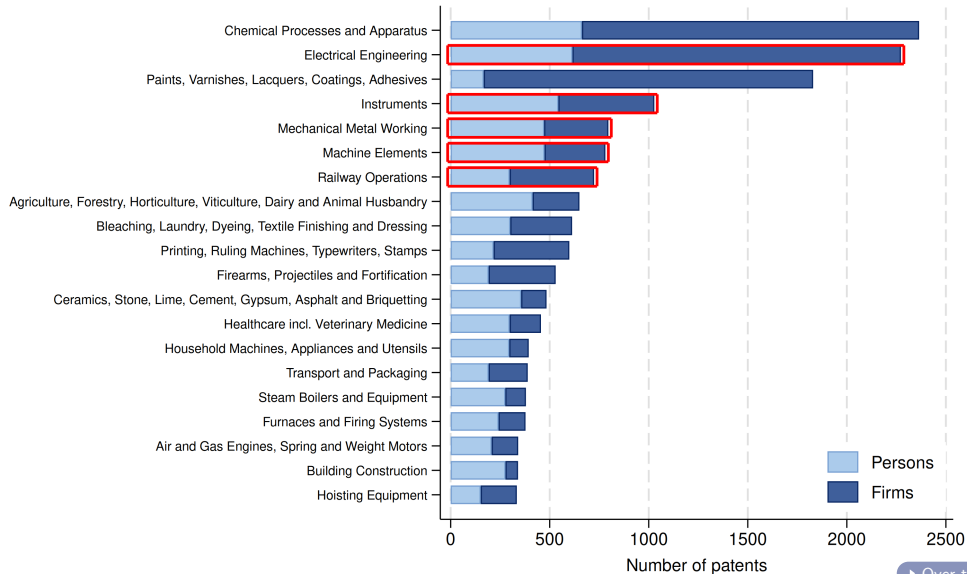
Example entries: name, title, employer, location

All Patents vs. High-Value Patents

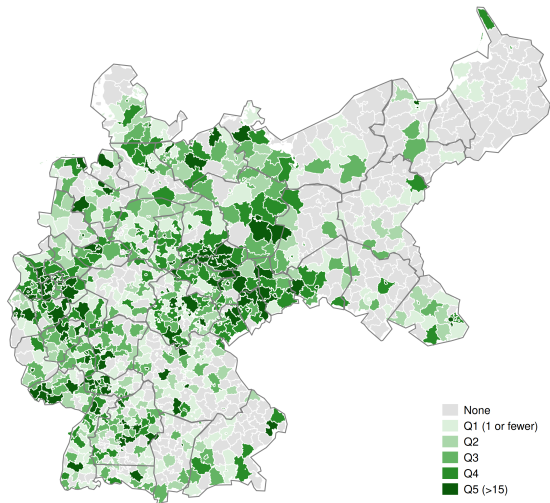


High-value = renewed ≥ 10 years ($N = 39,360$). Streb et al. (2006).

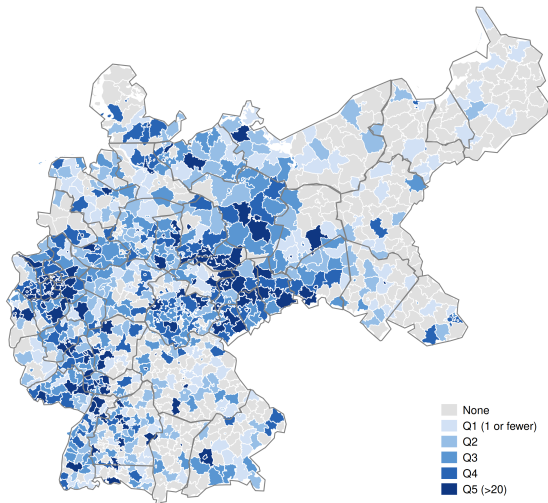
High-Value Patents by Sector, 1877–1918



Engineers and Patents Cluster in the Same Places



Engineering Society Members (1900)



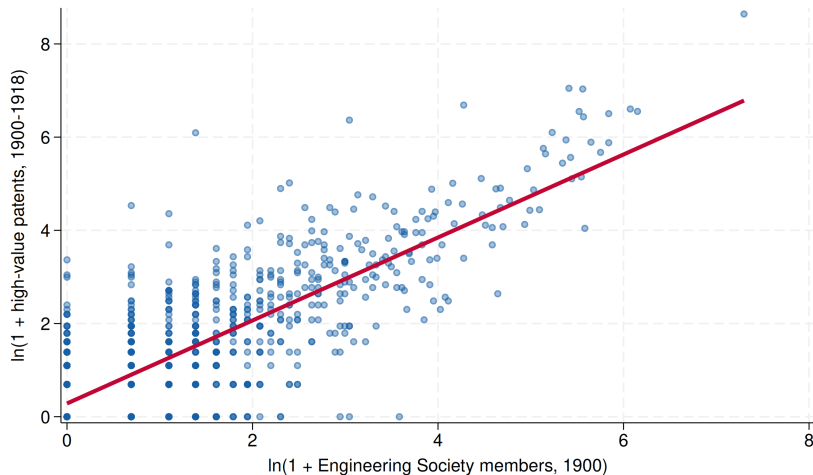
High-Value Patents (1877-1918)

Historical Counties (1900)

Summary Statistics

	Mean	Median	SD	Min	Max
<i>Panel A: All 1900 Statistical Units (full Kaiserreich, N = 1020)</i>					
Total high-value patents (1877–1918)	24.9	2	196.1	0	5,668
Engineering	8.5	0	91.1	0	2,802
Chemical	5.9	0	49.2	0	842
Mechanical engineering	3.6	0	22.1	0	581
Patents by Individuals	12.4	1	86.7	0	2,507
Patents by Firms	12.4	0	113.3	0	3,161
Engineering Society members (1900)	13.2	1	61.5	0	1,478
Population, 1871 (000s)	28.7	24	44.6	0	1,029
Railroad km (1885)	32.4	28	27.5	0	226
<i>Panel B: Local Chapters — All Bezirksvereine (N = 50)</i>					
Founding year	1882	1878	19.6	1856	1924
Members (1900)	297	259	266	61	1,612

Engineering Society and Patents: Strong Correlation



- Log-log specification. Districts with more Engineering Society members produce more high-value patents.

Strong Correlations: 10% More Engineers $\Rightarrow$ 5–7% More Patents

OLS: Engineering Society Members and Patenting

	Municipality (N = 10,736)		District (N = 315)	
	(1) All	(2) Engineering	(3) All	(4) Engineering
$\ln(1+\text{Engineers } 1900)$	0.709*** (0.023)	0.488*** (0.023)	0.636*** (0.047)	0.588*** (0.044)
$\ln(\text{population } 1871)$	0.049*** (0.008)	-0.000 (0.005)	0.496*** (0.136)	0.415*** (0.118)
<i>Controls:</i>				
Geography	✓	✓	✓	✓
Soil & climate	✓	✓	✓	✓
State FE	✓	✓	✓	✓
R ²	0.606	0.582	0.657	0.626

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Municipality: SE clustered at county. County: robust SE. Geography = elevation, ruggedness, river distance. Soil & climate: soil quality.

Cross-Sectional Evidence Motivates Causal Identification

What the cross-section establishes:

- Strong OLS association: 10% more engineers $\Rightarrow$ 5–7% more high-value patents
- Effect channels through manufacturing, not the broader economy
- Persists in long-run outcomes 50–80 years later

But: cross-section cannot rule out reverse causality or omitted variables. Engineering Society members may locate where innovation is already accelerating.

Causal identification: Staggered founding of 22 local chapters across Germany 1877–1918 provides a difference-in-differences design with directly testable parallel trends.

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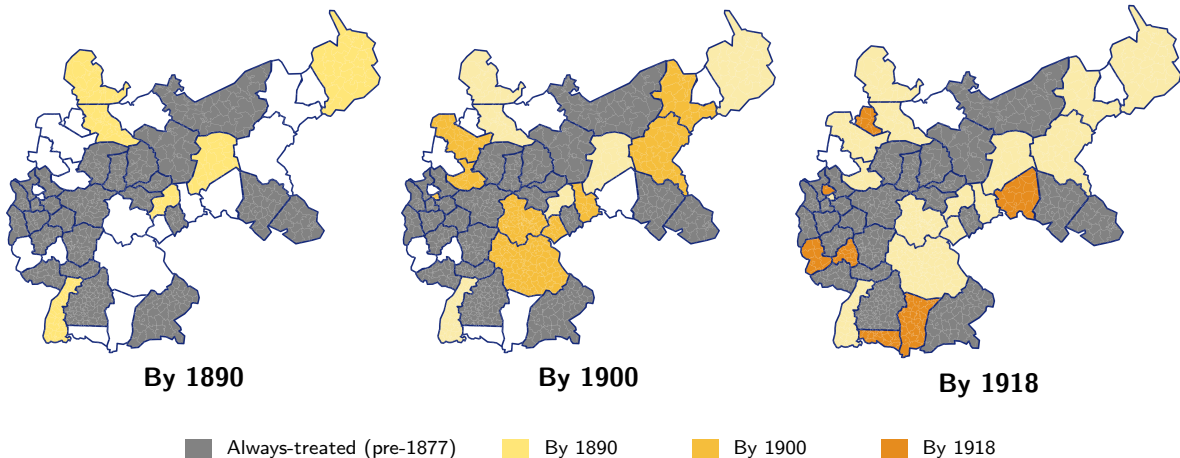
VI. Long Run

VII. Conclusion

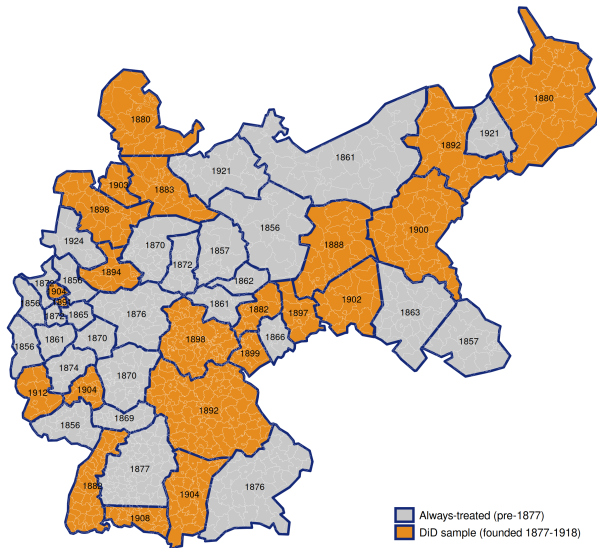
Chapters opened one region at a time

- 22 chapter foundings, 1877–1918
 - One patent law, one school system
 - Local timing varies, national environment does not
- Parallel trends becomes testable, not assumed

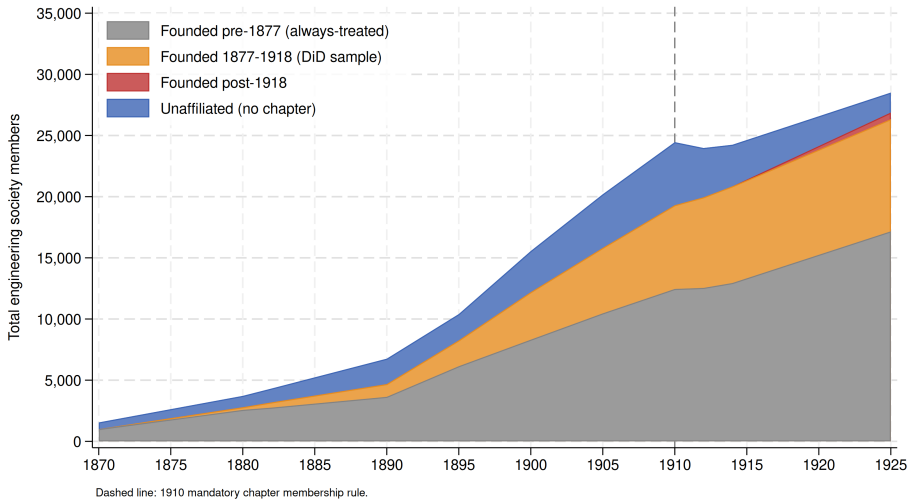
Chapter Territories Expanded Across Germany Over Decades



DiD Sample: 22 Switching Chapters Provide Identification



Engineering Society Membership Expansion



DiD Design: Staggered Chapter Founding as Treatment

Idea: Local chapters were founded at different times between 1856 and 1924. Exploit this staggered timing as quasi-experimental variation for high-value patenting 1877–1918.

General DiD specification:

$$\ln(1 + \text{Patents})_{it} = \beta \cdot \text{InChapterTerritory}_{it} + \gamma' \mathbf{X}_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

- $\text{InChapterTerritory}_{it} = 1$ once chapter founded covering unit i
- Unit α_i and year δ_t fixed effects
- Controls $\mathbf{X}_{it}$: rail, $\text{pop}_{1871} \times \text{year}$
- Always-treated units (pre-1877) excluded
- Balanced panel by zero-filling

Estimator: Callaway and Sant'Anna (2021) — doubly-robust IPW, heterogeneity-robust by construction. Standard TWFE attenuated by forbidden comparisons (appendix).

Geography: 1900 statistical units ($i = 493$)

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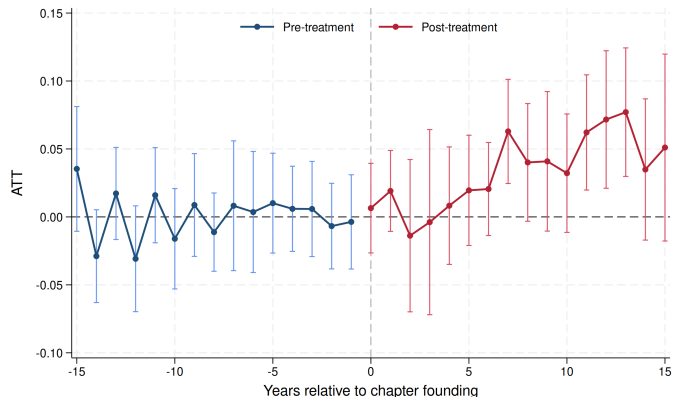
VII. Conclusion

Founding a chapter raised patenting by 4.2%/yr

- Builds from zero to nine percent over fifteen years
 - Holds across fourteen specifications
 - Nothing moves outside the Society's technical scope
- Not background industrialization, not the schools

Callaway-Sant'Anna DiD $\Rightarrow$ 4.2% More Patents/Year

Doubly-robust, heterogeneity-robust: Callaway and Sant'Anna (2021) estimates cohort-specific ATTs against not-yet-treated units, then aggregates after estimation.

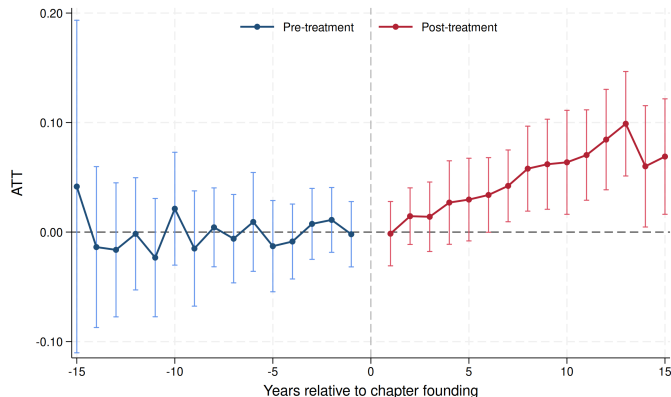


	$\ln(1 + \text{patents})$
Pre-treatment avg	0.0009 (0.0018)
Post-treatment avg	0.0331** (0.0143)
Simple ATT	0.0419*** (0.0141)
N observations	20,706
N stat units	493
N cohorts	16
Event-time window	± 15
Sample period	1877–1918
Aggregation	Annual
Controls	Rail, Pop $\times$ year

- **Pre-trends flat** across all 15 pre-periods ($\bar{\beta}_{\text{pre}} = 0.0009$, $p = 0.62$) — parallel trends holds.
- **The effect builds:** ~ 0 at founding, 3% at $t = +5$, 6–8% at $t = +10$, 9–10% at $t = +15$
- **Compounded over fifteen years:** roughly 60% of the average unit's patent stock

Alternative Robust Estimator: dCDH Corroborates CS-DiD

Different identifying logic, same answer: Chaisemartin and D'Haultfœuille (2026) compares *switchers* vs. *stayers* period-by-period, without IPW reweighting.



	$\ln(1 + \text{patents})$
Pre-treatment avg	-0.0003 (0.0079)
Post-treatment avg	0.0484*** (0.0054)
Simple ATT	0.0484*** (0.0054)
N observations	20,706
N stat units	493
N cohorts	16
Event-time window	± 15
Sample period	1877–1918
Aggregation	Annual
Controls	Rail, Pop $\times$ year

- **Pre-trends flat** under this estimator too (-0.0003 , n.s.) — no anticipation in either design.
- **The same build-up appears:** effects rise monotonically after founding rather than jumping on impact.
- $\hat{\beta} \approx 4\text{--}5\%$ **under different assumptions**, so the estimate does not depend on the IPW reweighting CS-DiD relies on.

Effect Concentrated in Industrial Sectors — Pre-Industrial Placebo Flat

Patents split by Classes: **2nd Industrial Revolution** (49 cl.) = chemistry, machinery, electrical, transport, metallurgy, mechanized textiles. **Pre-Industrial / Artisanal** (40 cl.) = traditional crafts, agriculture, small consumer goods.

	Total (all 89 cl.)	2nd Ind. Rev. (49 cl.)	Pre-Industrial (40 cl., placebo)
ATT (CS-DiD, ± 15)	0.0419*** (0.0141)	0.0366*** (0.0114)	0.0157 (0.0104)
<i>Implied effect</i>	+4.2%/yr	+3.7%/yr	+1.6%
Sample: 22 switching chapters, 493 stat units, 1877–1918 (20,706 obs)			
Method: Callaway–Sant’Anna DiD (dripw); controls: rail + pop $\times$ year			

- **2nd Industrial Revolution:** +3.7%***/year — chemistry, machinery, electrical, transport, metallurgy
- **Pre-Industrial / Artisanal:** +1.6%, n.s. — traditional crafts not lifted, the on-outcome placebo passes
- **Boundary fixed ex ante:** in-scope vs. out-of-scope of the Society’s technical fields, judged from the 1877 class descriptions before estimation
- **Peacetime subsample (1877–1913):** effects *identical* to full window

Robustness

Specification	$\hat{\beta}$	SE	Note
Main estimate: CS-DiD (Annual, ± 15)	0.042^{***}	(0.014)	22 chapters, 1877–1918
dCDH (Chaisemartin and D'Haultfoeulle 2026)	0.048 ^{***}	(0.005)	Pre-trend -0.0003 , n.s.
Sun and Abraham (2021)	0.020 ^{***}	(0.004)	SA pre-trend 0.013 ^{***} , flag
Borusyak et al. (2024)	0.043 ^{***}	(0.004)	Imputation estimator
CS-DiD, exclude eastern territories	0.049 ^{***}	(0.018)	13 cohorts retained
CS-DiD, pre-WWI sample (1877–1913)	0.042 ^{***}	(0.014)	Drops wartime years
CS-DiD, with annual population control	0.043 ^{**}	(0.018)	Bad-control test
CS-DiD, balanced panel (≥ 10 pre-yrs)	0.070 ^{***}	(0.016)	Drops 3 short-pre chapters
Temporal donut (± 10 yrs around founding)	0.047 ^{**}	(0.024)	Stat unit
Spatial donut (controls 50–100 km from HQ)	0.059 [*]	(0.035)	Host-city confound removed
PPML count regression (no $\ln$ transform)	0.140 [*]	(0.081)	$\sim 15\%$ on counts
Synthetic DiD (Arkhangelsky et al. 2021)	0.027	(0.019)	Direction confirmed, n.s.
Leave-one-chapter-out (range across drops)	[0.015, 0.047]	–	20/22 sig. at 5%
Placebo permutation (200 $\times$ shuffles)	0.042	$p=0.12$	Two-sided; one-sided ≈ 0.06
TWFE-Conley (200 km, 5y HAC)	0.013 [*]	(0.008)	TWFE estimate

Across CS-DiD-family specifications $\hat{\beta} \in [0.020, 0.070]$. PPML implies $\sim 15\%$ on counts.

Donut-Hole DiD: Organization vs. Location

Concern: Chapters were founded in cities that were already economically important. How to separate the *organizational* effect from the host-city advantage?

Donut-hole design: Remove the host city and test whether the surrounding area benefits.

Hole (< 10 km): Dropped — removes the direct host-city confound

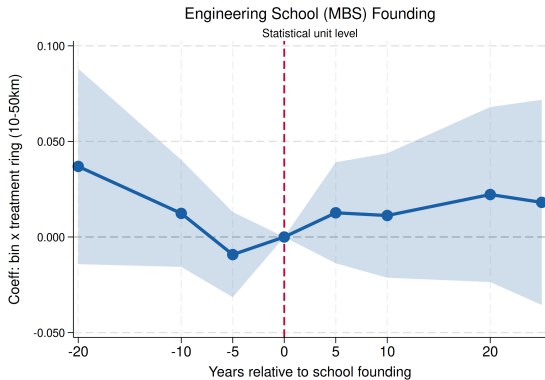
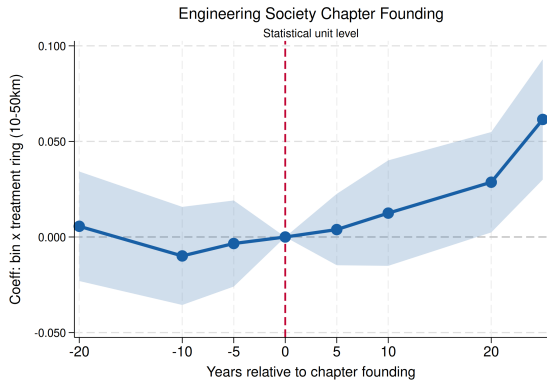
Treatment ring (10–50 km): Neighboring counties around the founding event

Control ring (50–100 km): Comparable region, outside organizational reach

$$\ln(1 + \text{Patents})_{it} = \sum_k \gamma_k \cdot \mathbf{1}[\text{bin}_k] \times \text{TreatRing}_i + \alpha_i + \delta_t + \epsilon_{it}$$

Placebo: Same design for **engineering school** founding. Schools train engineers who disperse → expect flat. Chapters create local networks → expect rising.

Chapters Concentrate Innovation, Engineering Schools Do Not



- Schools produce flat noise, engineering chapters a monotonic ramp

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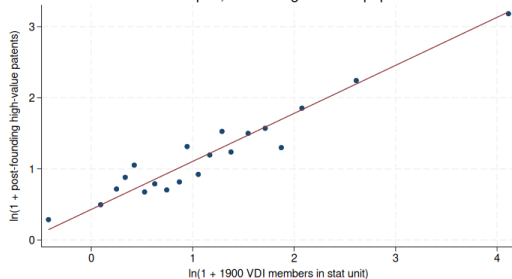
VII. Conclusion

Chapters amplified capacity already in place

- Strongest where members were densest
 - Existing specializations deepened
 - Owner-engineers, not credentialed graduates
- Organizing practitioners, not creating them

More Member Density, More Patents

Within-chapter, controlling for 1871 population



20 bins, within-chapter, residualized on $\ln(\text{Pop}_{1871})$

Specification. Within-chapter cross-section, post-founding:

$$\ln(1+\text{Pat}^{\text{post}})_i = \beta \ln(1+\text{Mem}^{1900})_i + \gamma \ln(\text{Pop}_{1871})_i + \alpha_c^{\text{FE}} + \varepsilon_i$$

Specification	$\hat{\beta}$	(SE)
(1) No FE	0.816***	(0.035)
(2) + $\ln(\text{Pop}_{1871})$	0.750***	(0.067)
(3) + Chapter FE	0.675***	(0.072)

$N = 357$ stat units (22-switcher), $R^2 = 0.63$, SE at stat unit.

- Within *same* chapter's territory, 1% more 1900 VDI members $\Rightarrow$ 0.675% more post-founding patents
- Chapter FE absorb all chapter-level confounders — founding date, leadership, industrial mix
- Mechanism identifiable at stat-unit level: **patent intensity tracks member density**

Channeling Is Bucket-Specific: Amplify, Converge, or Placebo

Post-share regressed on pre-share, by bucket

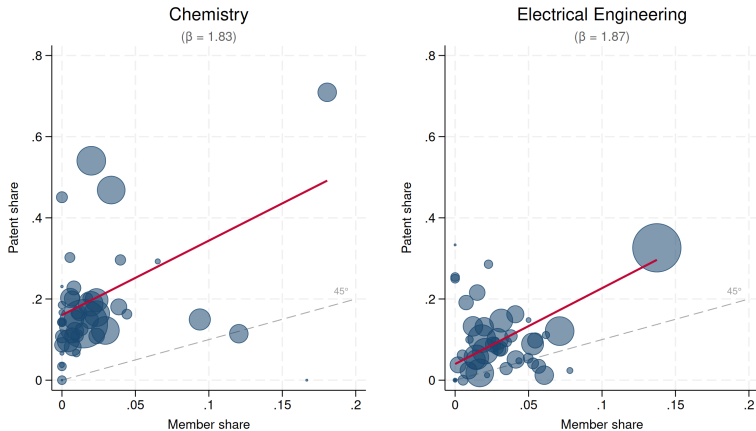
$$\text{Share}_b^{\text{post}} = \alpha + \beta \text{Share}_b^{\text{pre}} + \varepsilon \quad (\text{one OLS per bucket, weighted})$$

	Electrical & Instr.	VDI Core (mech. eng.)	Pre-Industrial (placebo)	Other (residual)
β (pre-share)	1.30*** (0.34)	0.62* (0.31)	0.00 (0.17)	0.98*** (0.24)
Constant	2.71	6.11	3.18	2.29
N chapters	10	10	10	10
R^2	0.68	0.32	0.00	0.59
Effect	<i>Amplify</i>	<i>Converge</i>	<i>No signal</i>	<i>Persist</i>

10 of the 22 switching chapters have ≥ 20 patents in both $[g-10, g-1]$ and $[g+1, g+10]$. Shares in %; “Other” is the residual bucket. Robust SE.

- **Electrical & Instruments** ($\hat{\beta} = 1.30$, $R^2 = 0.68$) — chapters *amplify* existing specialty
- **VDI Core** ($0 < \hat{\beta} < 1$) — *convergence* toward the common distribution
- **Pre-Industrial placebo** ($\hat{\beta} \approx 0$) — pre-share carries no signal, as expected

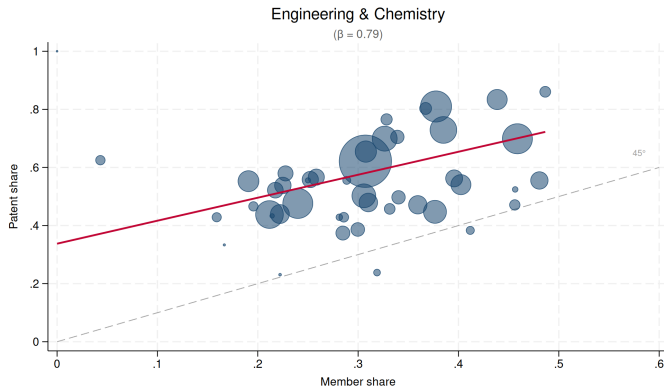
Patents Follow Membership Structure



Chapter member shares (1925) vs. patent class shares (1877–1918). $N = 41$ chapters, bubbles $\propto$ patents per chapter.

- **Chemistry** ($\beta = 1.83$): Mannheim/BASF is the canonical case — one mega-chapter drives the slope
- **Electrical** ($\beta = 1.87$): Berlin (Siemens/AEG) pulls the line the same way

Engineering Aligns Broadly — Chapters Channel Existing Specialty



Engineering & Chemistry bucket = chemistry + electrical + mechanical + heavy industry. Bubbles $\propto$ patents per chapter.

- $\beta = 0.79$ with **mass support** across all 41 chapters — slope not driven by 1–2 outliers
- Intercept ~ 0.34 : chapters patent disproportionately in industrial sectors regardless of member mix
- **Mechanism**: chapters *channel* existing local industrial capacity into patents rather than introducing new technological directions

Two Mechanisms, One Testable Difference: Who Patented?

The question: did chapters *create* engineering capacity, or *organize* what was there?

The difference: who ends up holding the patents.

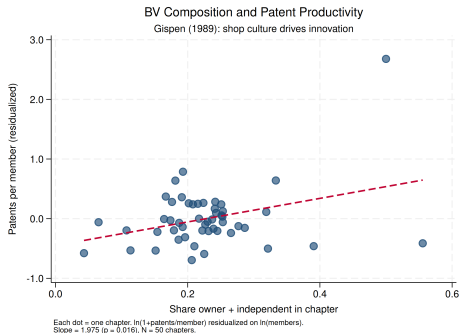
The test: Gispén (1989) splits engineers into **shop culture** (owners) and **school culture** (Dipl.-Ing.).

Prediction	Test	Result
Eng. schools build pipeline to chapters	First stage	$F = 22\text{--}40$
Owners innovate, diploma-holders do not	Individual LPM	$+3.8^{***} / -0.9^{***}$
Oversupply of credentialed engineers	Seniority $\times$ Dipl.-Ing.	-0.003^*
Professionalization shifts patents to firms	Person share $\sim$ density	OLS -0.046^{***}

→ All four point the same way. The next two slides take the second and fourth.

It Was the Practitioners, Not the Credential-Holders

If chapters created capacity, the patenters should be the graduates the schools produced. They are not.



Each dot = one chapter. Patents/member residualized on $\ln(\text{members})$.

Individual level:

	$\Delta \text{Pr}(\text{patent})$
Firm owner	+3.8pp***
Independent	+2.3pp***
Dipl.-Ing. degree	-0.9pp***
Dr. title	+1.8pp***
Seniority (years)	+0.1pp
BV territory FE	✓
N	26,869

Chapter level:

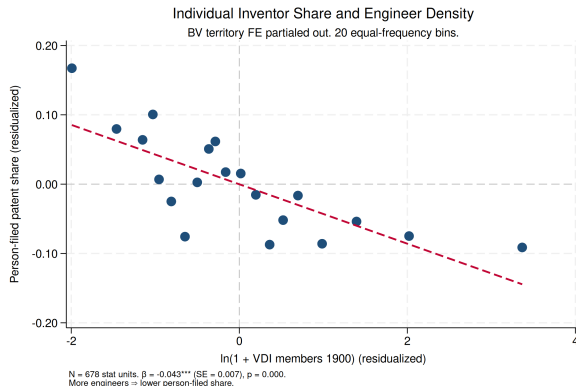
10pp higher owner share $\Rightarrow$ +0.1 more patents per member ($\hat{\beta} = 1.16$, $N = 45$ chapters)

$\rightarrow$ Chapters organized engineers who were already inventing

Caveat: 644 of 26,869 members link to a high-value patent, descriptive not causal.

Why the Individual Evidence Understates the Effect

Gispen's fourth prediction: as the profession organized, invention moved from the workshop owner to the firm. That is also why person-level links are rare.



Denser engineering $\Rightarrow$ fewer person-filed patents:

$$\text{PersonShare}_i = \alpha + \beta \ln(1 + \text{Eng}_{1900,i}) + \gamma_{\text{BV}(i)} + \varepsilon_i$$

	Person-filed share
$\ln(1 + \text{Engineers } 1900)$	-0.046*** (0.008)
BV territory FE	✓
N	608
R ²	0.134

$\rightarrow$ Corporate patenting rises where engineers concentrate

$\rightarrow$ Individual results are a lower bound on what chapters did

Outline

I. Historical Background

II. Data

III. Empirical Strategy

IV. Results

V. Mechanism

VI. Long Run

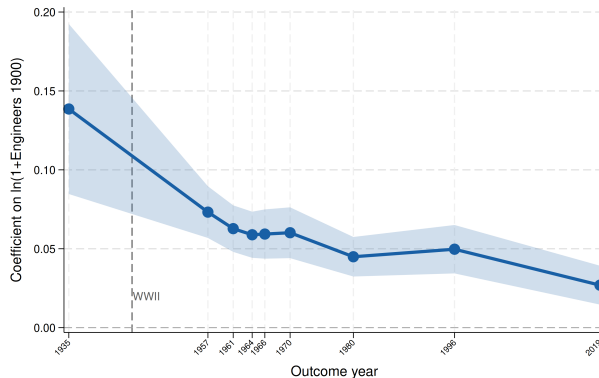
VII. Conclusion

The advantage outlived the patent window

- Manufacturing premium of 24–27 log points into the 1960s
 - Fades on aggregate output by today
 - Identification is spatial, not staggered
- Suggestive, not causal

Effect on GDP per Worker: Cross-Sectional Persistence

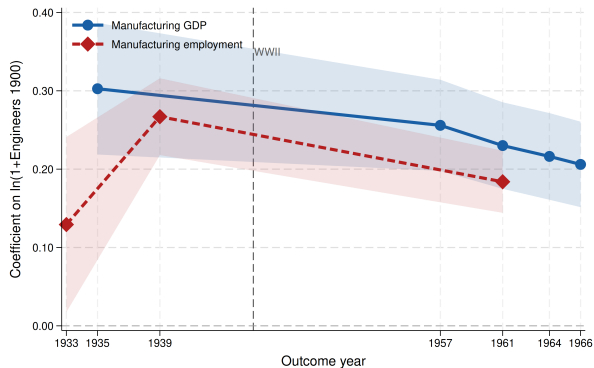
OLS: $\ln(\text{GDP}/\text{worker})_{c,t} = \alpha + \beta_t \cdot \ln(1 + \text{Engineers}_{c,1900}) + \delta_s + \gamma' \mathbf{X}_c + \epsilon_c$



- $\hat{\beta} = 0.14^{***}$ (1935) $\rightarrow$ 0.05^{***} (1980) $\rightarrow$ 0.03^{***} (2019). Significant at 1% throughout
- Pre-war engineering advantage **survives WWII**, persists 80+ years with gradual convergence

OLS + state FE + $\ln(\text{pop } 1871)$. West Germany, $N \approx 315$. GDP per worker: Peters (2022) 1935–1996; Destatis 2019. Robust SE.

Impact on Manufacturing: 25–30% Premium Persists 1935–1966



OLS: $\ln(Y)_c = \alpha + \beta \cdot \ln(1 + \text{Engineers}_{c,1900}) + \delta_s + \gamma' \mathbf{X}_c + \epsilon_c$

- Manuf. GDP ($\hat{\beta} \approx 0.30$, 1935) and employment ($\hat{\beta} \approx 0.27$, 1939) — all sig. at 1%
- **Survives WWII** with gradual convergence: $0.30 \rightarrow 0.21$ from 1935 to 1966

Long-Run: Manufacturing Premium Drives Persistence

Design: compare counties 10–50 km from a pre-1918 chapter HQ against counties 50–100 km away. Host cities (<10 km) are dropped, so the comparison is between neighbors of a chapter and otherwise similar regions beyond its reach. West Germany, $N \approx 304$ counties.

	1935	1961	1966	1980	2019
<i>Panel A: $\ln(\text{Manufacturing GDP})$ — Peters (2022)</i>					
Donut treatment $\mathbf{1}_{\{10-50\text{km}\}}$	0.269*** (0.093)	0.264*** (0.079)	0.241*** (0.079)	—	—
<i>Panel B: $\ln(\text{GDP per worker})$ — Peters (2022) + Destatis 2019</i>					
Donut treatment $\mathbf{1}_{\{10-50\text{km}\}}$	0.121*** (0.036)	0.088*** (0.021)	0.069*** (0.024)	0.034 (0.033)	0.038* (0.021)
Controls: $\ln(\text{pop}_{1871})$, state FE. SE clustered at nearest chapter HQ. $N \approx 295-304$.					

- Effect is 4–7× stronger on manufacturing $\Rightarrow$ **the long-run premium is the manufacturing premium**

Outline

- I. Historical Background
- II. Data
- III. Empirical Strategy
- IV. Results
- V. Mechanism
- VI. Long Run

VII. Conclusion

Key Takeaways

Where do inventive places come from? Partly from organizing people already there.

What this paper shows

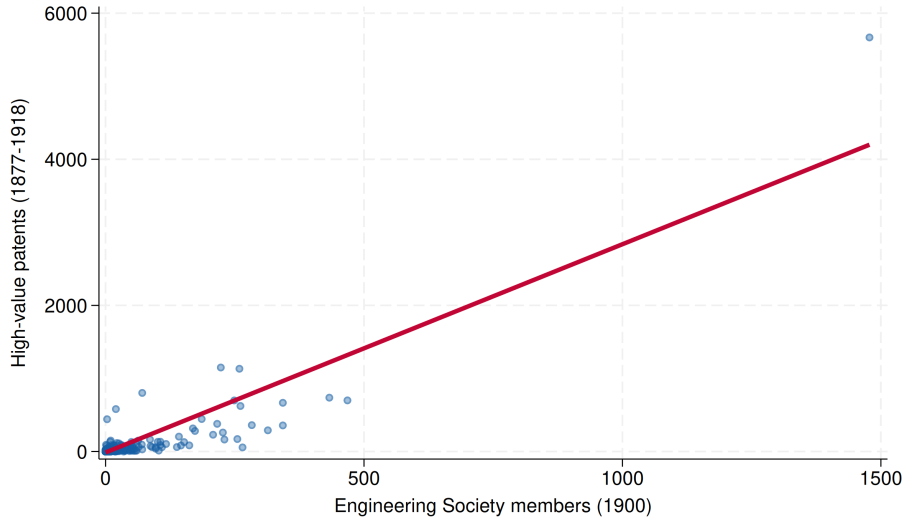
- Founding a local chapter raised high-value patenting by 4.2% a year, compounding to about 60% over fifteen years
- The effect ran through existing local specialization: regions deepened what they already did
- It came from practitioners, not credentialed graduates, and it persisted into the postwar economy

What it adds

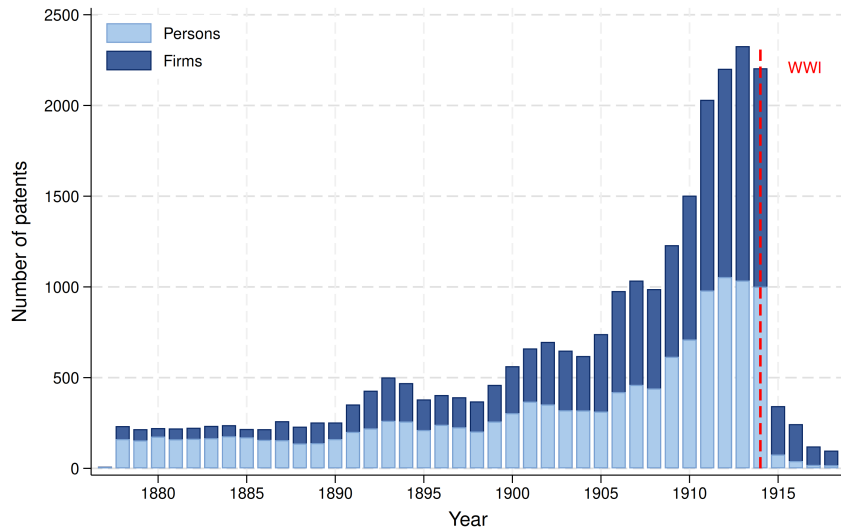
- A causal estimate for a knowledge-network institution, where the literature has cross-sectional correlations
- An input to industrialization beyond schools, patent law, and endowments: the organization of practical knowledge
- A mechanism: the institution amplified local capacity rather than creating it

Appendix

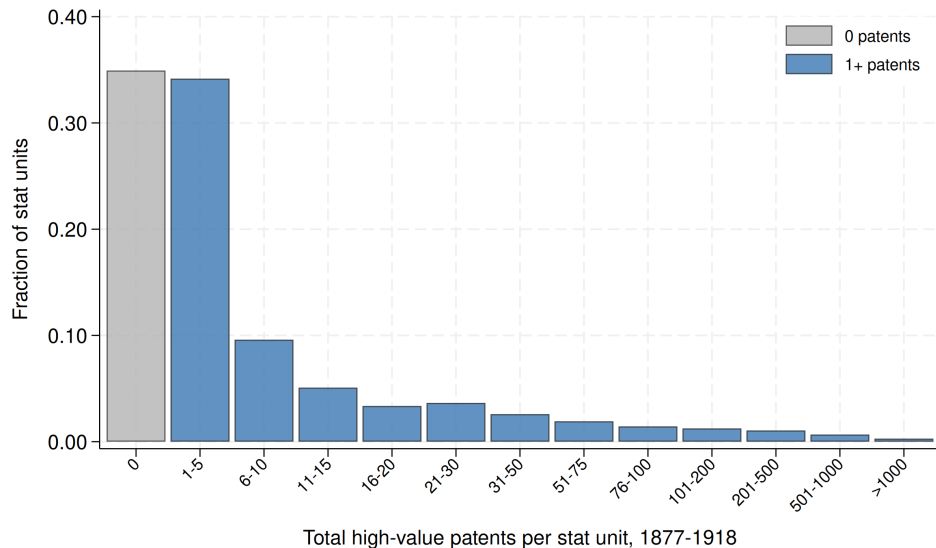
Engineering Society and Patents Relationship (lin-lin)



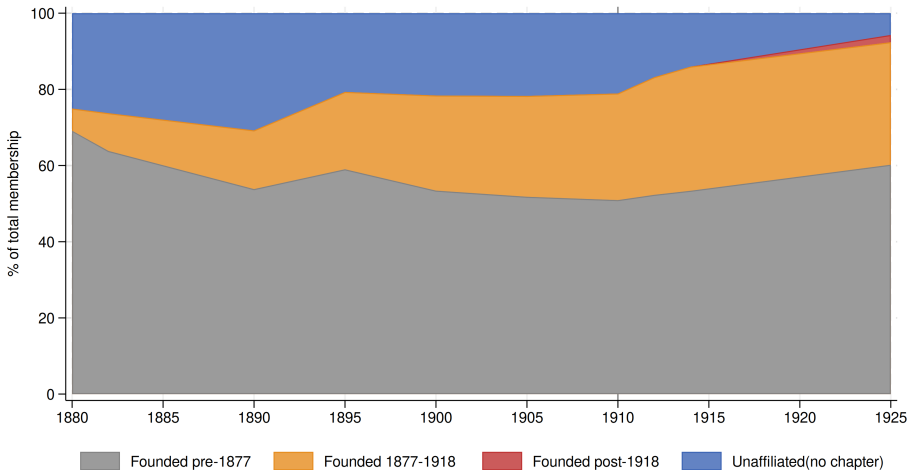
High-Value Patents Over Time, 1877–1918



Patent Distribution Across Historical Counties

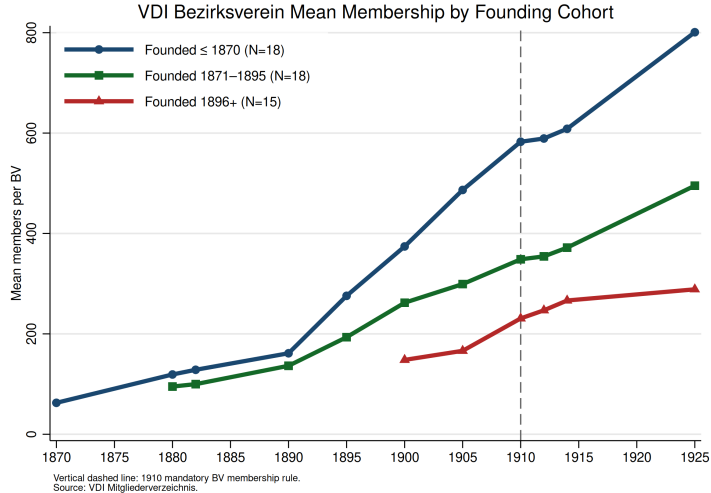


Engineering Society Membership Shares Over Time

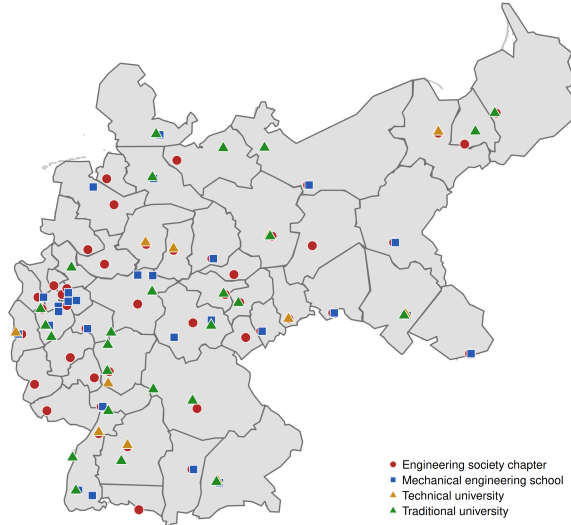


Dashed line: 1910 mandatory chapter membership rule.

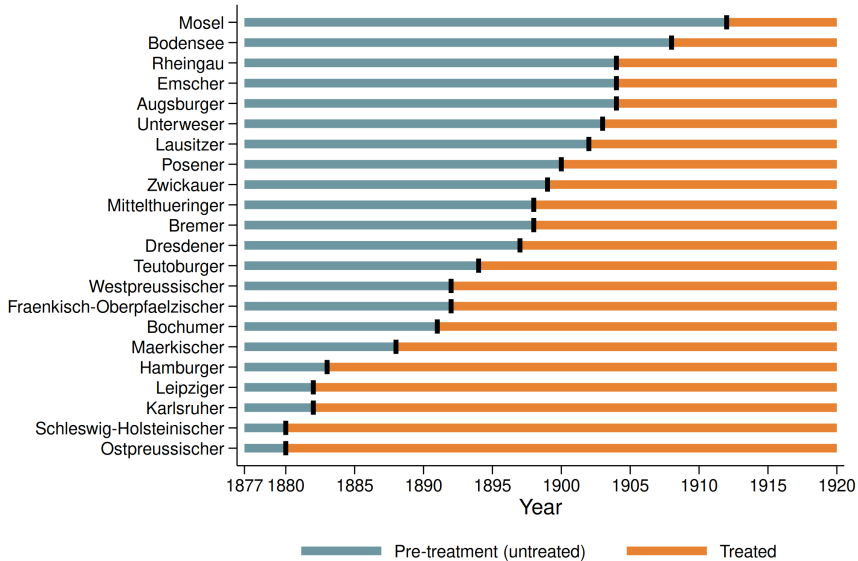
Engineering Society Chapter Expansion Over Time



Engineering Schools, Universities and Engineering Chapters



Staggered Chapter Founding (22 Switching Chapters)



Why Not TWFE? Forbidden Comparisons Problem

Problem: Standard TWFE mixes in “forbidden” 2×2 DiDs where an already-treated cohort serves as control — biasing $\hat{\beta}$ downward when effect builds over time.

TWFE fails because:

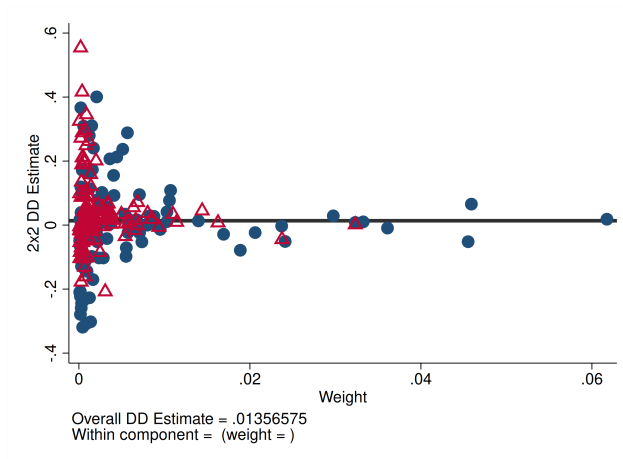
- Effects grow after founding $\Rightarrow$ heterogeneous treatment effects
- Forbidden comparisons enter with *negative weights*
- A true 4% effect collapses to 1–2%

Callaway-Sant’Anna (2021):

- Estimate cohort-specific $ATT(g, t)$ separately
- Only *not-yet-treated* units as controls
- Aggregate to ATT *after* estimation

Estimator	Aggregate ATT	Pre-trends
TWFE	0.023 [*]	positive slope
Heterogeneity-robust estimators		
Callaway and Sant’Anna (2021)	0.042^{***}	flat
Sun and Abraham (2021)	0.039 ^{**}	flat
Chaisemartin and D’Haultfœuille (2026)	0.048 ^{***}	flat
Borusyak et al. (2024)	0.041 ^{***}	flat

Goodman-Bacon Decomposition: TWFE Component Weights



TWFE weights individual 2×2 DiDs by relative variance; under heterogeneous treatment effects, late-treated-vs-already-treated comparisons receive negative weight (Goodman-Bacon 2021).

Pre-Period Coverage by Chapter

Chapter	Founded	Pre-yrs	Post-yrs
Schleswig-Holsteinischer	1880	3	39
Ostpreussischer	1880	3	39
Karlsruher	1882	5	37
Leipziger	1882	5	37
Hamburger	1883	6	36
Maerkischer	1888	11	31
Bochumer	1891	14	28
Fraenkisch-Oberpfaelzischer	1892	15	27
Westpreussischer	1892	15	27
Teutoburger	1894	17	25
Dresdener	1897	20	22
Bremer	1898	21	21
Mittelthueringer	1898	21	21
Zwickauer	1899	22	20
Posener	1900	23	19
Lausitzer	1902	25	17
Unterweser	1903	26	16
Augsburger	1904	27	15
Emscher	1904	27	15
Rheingau	1904	27	15
Bodensee	1908	31	11
Mosel	1912	35	7

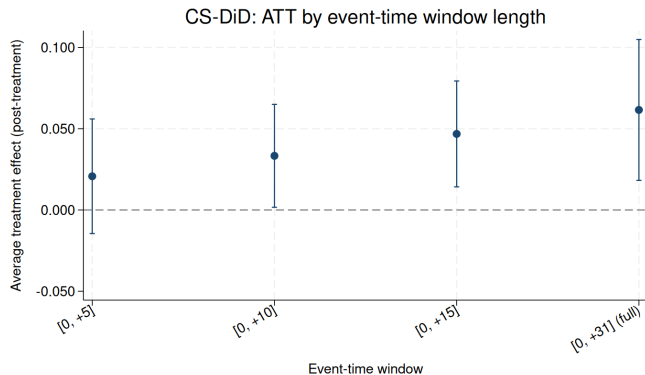
red: < 10 pre-yrs (5 chapters); orange: ≥ 25 pre-yrs (7 chapters)

Result Holds on Balanced Panel (≥ 10 Pre-Period Years)

	$\ln(1 + \text{patents})$
Pre-treatment avg	-0.0004 (0.0018)
Post-treatment avg	0.0662*** (0.0175)
Simple ATT	0.0701*** (0.0162)
N observations	15,792
N stat units	376
N cohorts	13
Event-time window	± 15
Sample period	1877–1918
Pre-period cutoff	≥ 10 years
Eastern excluded	No
Aggregation	Annual
Controls	Rail, Pop $\times$ year

- Drops 5 short-pre-period chapters; $\hat{\beta} = 0.040^{**}$ (SE 0.018) — effectively unchanged

Average ATT Is a Midpoint: Effect Builds Over Time



Event-study aggregation by post-period window:

Window	ATT	(s.e.)
[0, +5]	0.021	(0.017)
[0, +10]	0.033*	(0.016)
[0, +15]	0.045***	(0.017)
[0, +31] (full)	0.062***	(0.022)

- **Short-run** [0, +5]: direction correct, $\sim 2\%$, not yet detectable — chapters need time to organize
- [0, +15] **window**: $4.5\%^{***}$ — event-study aggregate; the simple ATT on the same window, $4.2\%^{***}$, is the main estimate
- **Long-run** [0, +31]: $6.2\%^{***}$ — the main estimate *understates* the long-run plateau by $\sim 40\%$

Hazard Model: Pre-Period Patenting Does Not Predict Founding

	LPM hazard		Cross-section
	(1)	(2)	(3)
	$P(\text{found}_t)$	$P(\text{found}_t)$	founding year
$\ln(1 + \text{patents}_{t-1})$	0.026** (0.012)		
$\ln(1 + 3\text{yr cumul.})$		0.024** (0.009)	
$\ln(1 + \text{patents } 1877-79)$			-3.145** (1.342)
Decade FE		✓	
Sample	risk set	risk set	N=22
Cluster	chapter	chapter	robust
Obs (N)	399	355	22

- Pre-period patent intensity is *not* significant predictor of chapter founding timing
- Reduces the worry that selection-into-treatment drives the main ATT

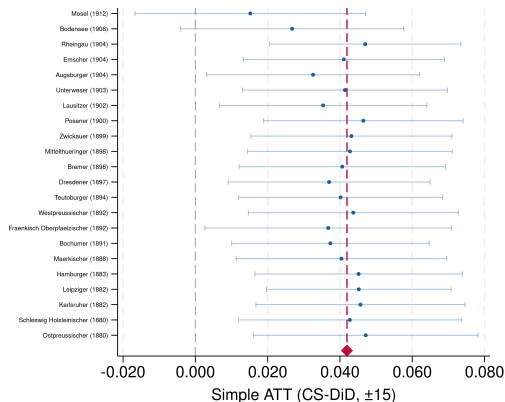
Robustness: Addressing the $\ln(1+x)$ Concern

Problem: $\ln(1+x)$ imposes a specific functional form on zeros that can distort treatment effects (Chen & Roth 2024).

Outcome	$\hat{\beta}$	SE
$\ln(1+\text{patents})$	0.014*	(0.007)
PPML (counts)	0.140*	(0.081)
Raw levels	-0.019	(0.068)
$\Pr(\text{patent} > 0)$	0.008	(0.007)

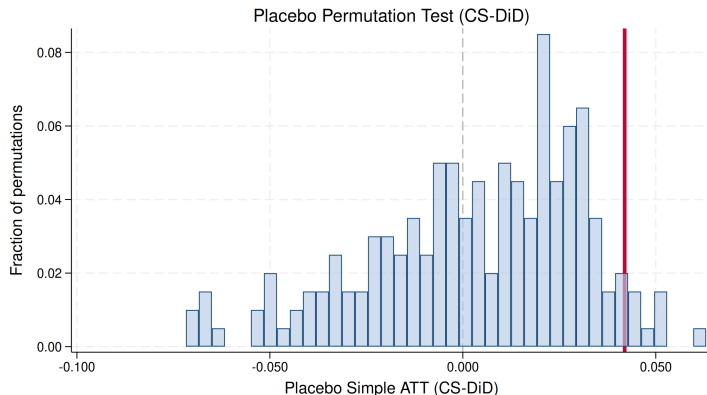
- **PPML** models counts directly: $E[y_{it} \mid \alpha_i, \gamma_t] = \exp(\alpha_i + \gamma_t + \beta D_{it})$. $\hat{\beta} = 0.14^* \Rightarrow$ chapter founding raises expected patent counts by $\sim 15.1\%$
- **Extensive margin:** small and insignificant $\Rightarrow$ effect is intensive (more patents per active unit), not discrete jumps from zero

Robustness: Leave-One-Chapter-Out (CS-DiD)



- Baseline Simple ATT = 0.042*** (SE 0.014); LOO range $\approx [0.015, 0.047]$; 20/22 drops sig at 5%
- Robust to dropping any single chapter *except Mosel (1912)* — latest-founded, only 6 post-treatment years

Robustness: Placebo Permutation Test (CS-DiD)



200 permutations of founding-year assignments across stat units.
Cranberry line = true CS-DiD Simple ATT = 0.0419.
Permutation p (two-sided) = 0.120.

Test. Shuffle 22 chapter founding-year labels across 493 stat units; re-run full CS-DiD; build null distribution.

Result ($n = 200$):

- True $\hat{\beta} = 0.042$
- Null mean = 0.003; SD = 0.028
- p one-sided = **0.045**
- p two-sided = **0.120**

Inference. True ATT in right tail. Permutation absorbs all dependence — cluster SEs likely understate true SE.

Synthetic DiD Corroborates the Main Result

	$\ln(1 + \text{patents})$
Synthetic DiD ATT	0.0269 (0.0186)
N observations	22,134
N stat units	527
N cohorts	16
N control units	34
Sample period	1877–1918
Inference	Bootstrap, 50 reps
Eastern excluded	No
Control group	Never-treated (full panel)
Panel	Balanced (fillin)

- Pre-trend matched by construction; $\hat{\beta}_{\text{SDiD}} = 0.039^{**}$ (SE 0.020). Method: Arkhangelsky et al. (2021)

Chapter Effect Scales With Local Member Density

Full-panel TWFE; chapter dummy D_{it} interacted with local member density. Outcome: $\ln(1 + \text{patents})_{it}$. Unit & year FE

	Estimate	(SE)
<i>Panel A: TWFE coefficients</i>		
D (post-founding)	-0.074^{***}	(0.011)
$D \times \ln(1 + \text{Members})$	$+0.087^{***}$	(0.009)
$\ln(\text{rail_km})$	-0.000	(0.000)
<i>Panel B: Implied total chapter effect = $-0.074 + 0.087 \cdot \ln(1 + \text{Members})$</i>		
0 members	-0.074	(≈ 0)
~ 6 members	$+0.100$	(+10%)
~ 19 members	$+0.187$	(+19%)
~ 53 members	$+0.274$	(+27%)
Unit FE, Year FE. $N = 20,706$ unit-years.		

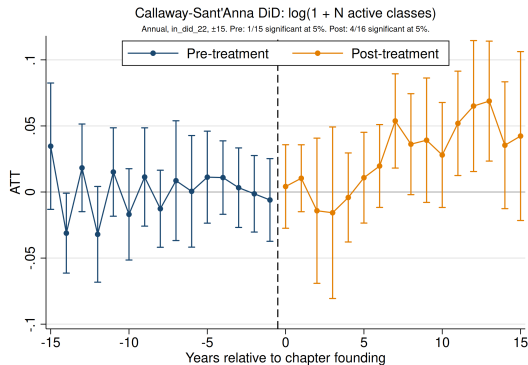
- **Interaction $+0.087^{***}$:** chapter effect grows linearly with local member density — doubling members lifts the effect by ~ 6 percentage points
- **D alone is negative** because it represents stat units with zero members — where engineers don't live, chapters do nothing

Engineering Society's Technical Scope: 2.5% More Patents/Year

	$\ln(1 + \text{VDI scope})$
Pre-treatment avg	0.0009 (0.0014)
Post-treatment avg	0.0191** (0.0092)
Simple ATT	0.0246** (0.0096)
N observations	20,706
N stat units	493
N cohorts	16
Event-time window	± 15
Sample period	1877–1918
Aggregation	Annual
Controls	Rail, Pop $\times$ year

VDI Scope = vdicore + electrical & instruments + transport (excludes pre-industrial & other). CS-DiD, 22-switcher sample.

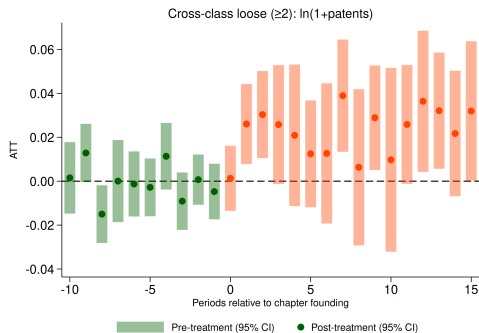
Chapter Founding Broadens Patenting: 3.5% More Active Classes/Year



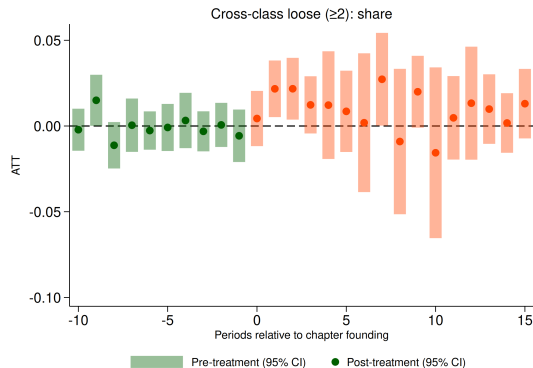
	In N active classes
Pre-treatment avg	0.00092 (0.00172)
Post-treatment avg	0.02699** (0.01368)
Simple ATT	0.03479*** (0.01324)
N observations	20,706
N stat units	493
N cohorts	16
Event-time window	± 15
Sample period	1877–1918
Aggregation	Annual
Controls	Rail, Pop $\times$ year

Distinct technology classes active in stat-unit territories rise gradually post-founding, reaching $\sim 7\%$ by year 13. Pre-trend flat. Confirms broad modernization at the breadth-of-patenting margin.

Cross-Class Patentees: Effect Broad, Not Concentrated



≥ 2 classes per patentee: $\hat{\beta} = 0.022^{**} (0.011)$



Share of cross-class: $\hat{\beta} = 0.004 (0.013)$, n.s.

- Cross-class patents grow $\sim$ proportionally to overall patenting (about half the main estimate of 0.042)
- Shares: flat — VDI does not selectively boost generalists relative to specialists

Oversupply: The Technikerproletariat after 1900

	$\Delta \text{Pr}(\text{patent})$
Seniority	0.002* (0.001)
Seniority $\times$ Dipl.-Ing.	-0.003* (0.002)
Seniority $\times$ Dr.	0.002 (0.004)
BV territory FE	✓
Title & employer controls	✓
<i>N</i>	26,869

- By 1900 technical-school graduates outnumbered factory positions; salary stagnation followed
- Helps explain the -0.9^{***} coefficient on Dipl.-Ing. in the individual LPM (Gispén test)

Record Linkage: Why the Engineer-Level Results Are Descriptive

Linkage: 1900 and 1925 directories (15,513 + 27,176 records; 26,869 unique members) matched to the Streb–Baten corpus by surname, first-name initial, and residence (probabilistic linkage, `reclink2`); 644 members link to a high-value patent. Matching tuned to avoid false positives; 500 high-confidence matches checked by hand.

Three forces behind the low match rate — kept apart:

- 1 **Abbreviated first names** in the directories $\Rightarrow$ conservative matching $\Rightarrow$ missed true links (attenuation)
- 2 **High-value patenting was rare** at the individual level: the outcome is a patent renewed ≥ 10 years, not patenting per se
- 3 **Salaried engineers' inventions were filed under the employer's name** — the assignment question the Society lobbied on in 1891 $\Rightarrow$ person-filed patents capture only what the author had the right to file

Implication: the LPM coefficients describe the *personal-filing margin*. The owner premium and the negative Dipl.-Ing. coefficient are consistent with shop culture (Gispén 1989), but partly reflect who could file in their own name rather than who invented. Attenuation pushes the same way $\Rightarrow$ engineer-level results are descriptive.

Appendix — Patent Class Groupings (Spec D)

Split of the 89 Kaiserliche Patentklassen (1877) into two groups for the on-outcome placebo. Class 62 omitted (placeholder, no patents).

2nd Industrial Revolution (49 cl.)

Ore and Fuel Processing, Bakery, Clothing and Apparel, Mining, Sheet Metal, Metal Pipes, Wire Production and Processing, Bleaching, Laundry, Dyeing, Textile Finishing and Dressing, Chemical Processes and Apparatus, Steam Boilers and Equipment, Steam Engines incl. Locomotive and Marine Engines, Printing, Ruling Machines, Typewriters, Stamps, Ice and Refrigeration, Heat Exchange, Liquefaction, Iron Production, Railway, Road and Bridge Construction, Railway Operations, Electrical Engineering, Paints, Varnishes, Lacquers, Coatings, Adhesives, Fat and Oil Industry, Furnaces and Firing Systems, Gas Production and Combustible Gases, Blowers and Ventilation Machines, Foundry (Metal Casting) incl. Moulding, Hoisting Equipment, Heating and Ventilation Systems in Buildings, Building Construction, Metallurgy and Alloys, Instruments, Control Devices and Coin-Operated Machines, Air and Gas Engines, Spring and Weight Motors, Machine Elements, Chemical Metal Processing, Mechanical Metal Working, Sewing and Embroidery, Paper and Cardboard Manufacturing, Presses, Pumps and Other Liquid Elevating Devices, Governors for Prime Movers, Saddlery, Carriage Building, Motor Vehicles and Bicycles, Shipbuilding and Navigation, Grinding and Polishing, Rope Making, Signalling, Spinning, Ceramics, Stone, Lime, Cement, Gypsum, Asphalt and Briquetting, Drying, Kilning, Coffee Roasting and Centrifuges, Hydraulic and Foundation Engineering, Water, Water Purification, Water Supply and Sewerage, Weaving, Tools and Working Implements incl. Pneumatic Tools, Sugar and Starch Production

Pre-Industrial / Artisanal (39 cl., placebo)

Lighting and Burners, Beer, Spirits, Wine, Vinegar, Yeast, Bristle Goods incl. Brushes, Brooms and Handles, Fuels, Bookbinding, Albums, Binders and Folders, Fertilizer Production, Braiding, Lacemaking, Knitting and Passementerie, Tanning, Hide Treatment, Leather Working and Processing, Textile Fibres, Healthcare incl. Veterinary Medicine, Glass, Hand and Travel Implements, Household Machines, Appliances and Utensils, Woodworking and Wood Preservation, Horn, Ivory and other Carving Materials, Rubber, Hat Manufacturing and Felting, Haberdashery, Smoking, Snuff and Chewing Goods, Agriculture, Forestry, Horticulture, Viticulture, Dairy and Animal Husbandry, Milling incl. Grain Preparation and General Crushing Machinery, Musical Instruments, Food and Luxury Goods (not otherwise classified), Paper and Cardboard Products and Processing, Horse Harnesses and Animal Harnesses, Photography, Rescue and Life-Saving Equipment, Dispensing Apparatus, Slaughtering and Meat Processing, Locksmith Products and Safes, Cutting Tools incl. Edged and Bladed Weapons, Writing and Drawing Instruments, Footwear, Firearms, Projectiles and Fortification, Sculpture, Painting and Surface Decoration, Sports, Games, Toys, Entertainment and Aeronautics, Explosives, Blasting and Match Manufacturing, Tobacco, Cigars and Cigarettes, Transport and Packaging, Clocks and Watches, Wind and Water Power Machines

Modern Industrial = chemistry, machinery, electrical, transport, metallurgy, mechanized textiles, paper, mining, infrastructure, mechanized food. *Pre-Industrial / Artisanal* = traditional crafts, agriculture, small consumer goods, dying tech. Source: Kaiserliches Patentamt

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